

Introduction to Virtualization Features

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Foreword

- Having attended previous courses about cloud computing, you may already have a good understanding of virtual machines (VMs) and the resources they use. This chapter focuses on when and where we would need to use VMs and what virtualization features VMs provide.

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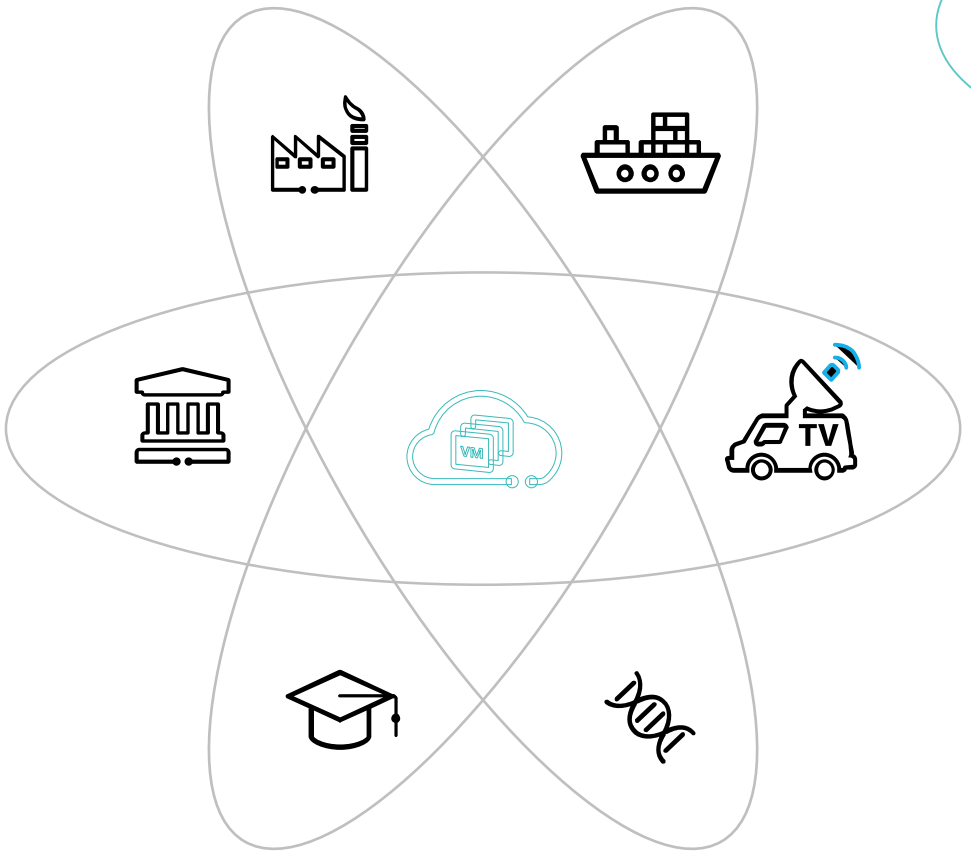


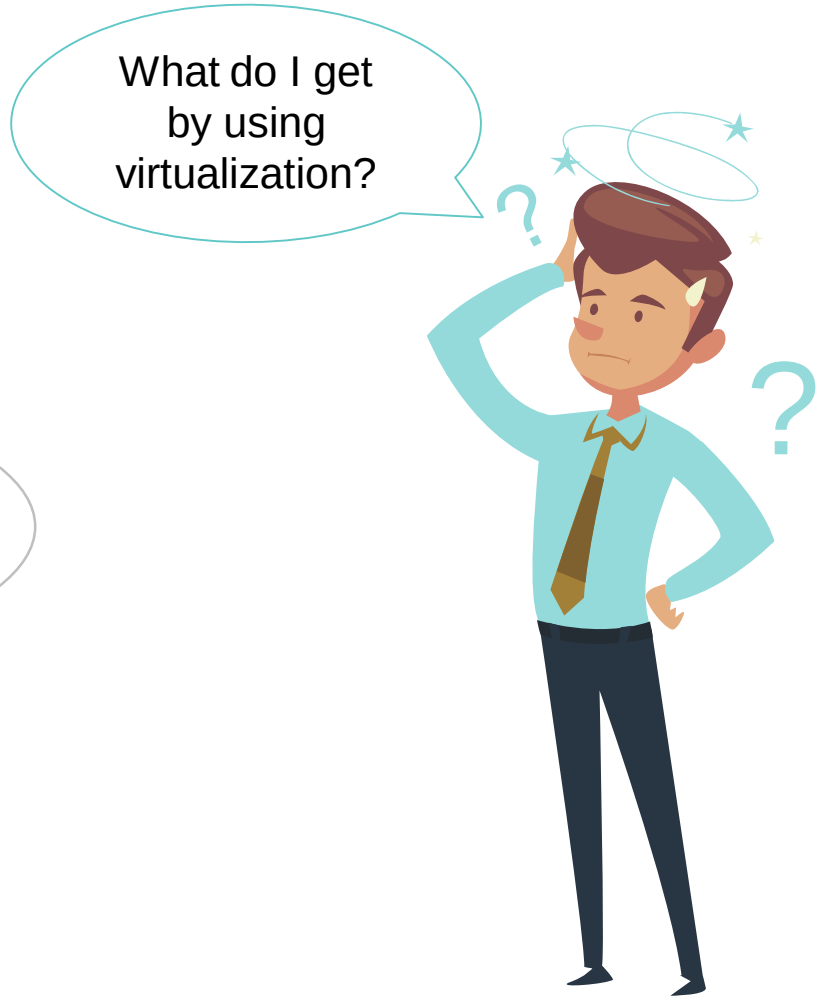
Objectives

- Upon completion of this course, you will:
  - Understand virtualization features.
  - Understand where and when virtualization features may be used.

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What Are the Benefits of Virtualization?





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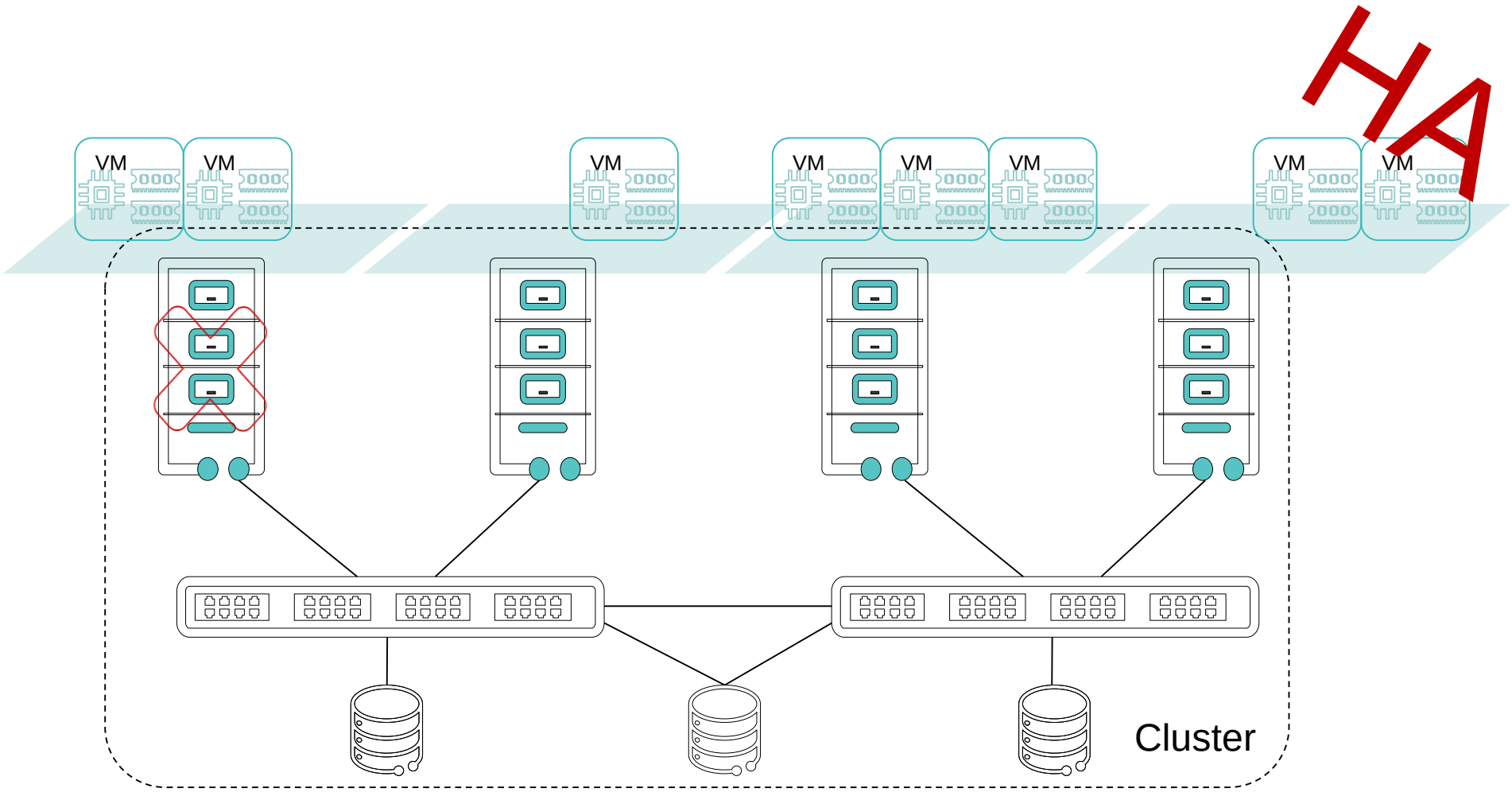
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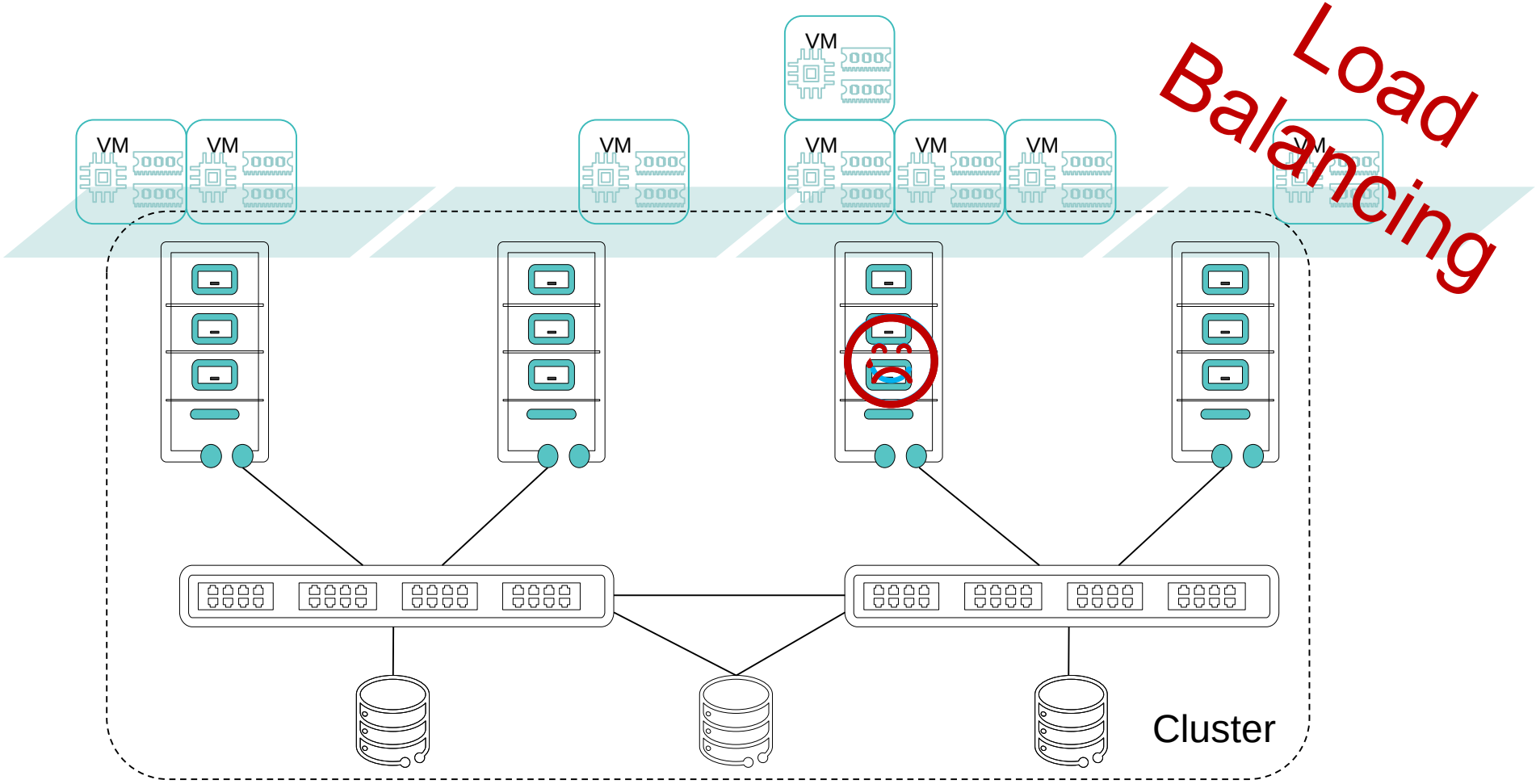
Virtual Cluster Features (1)





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Virtual Cluster Features (2)



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Virtual Cluster Features (3)

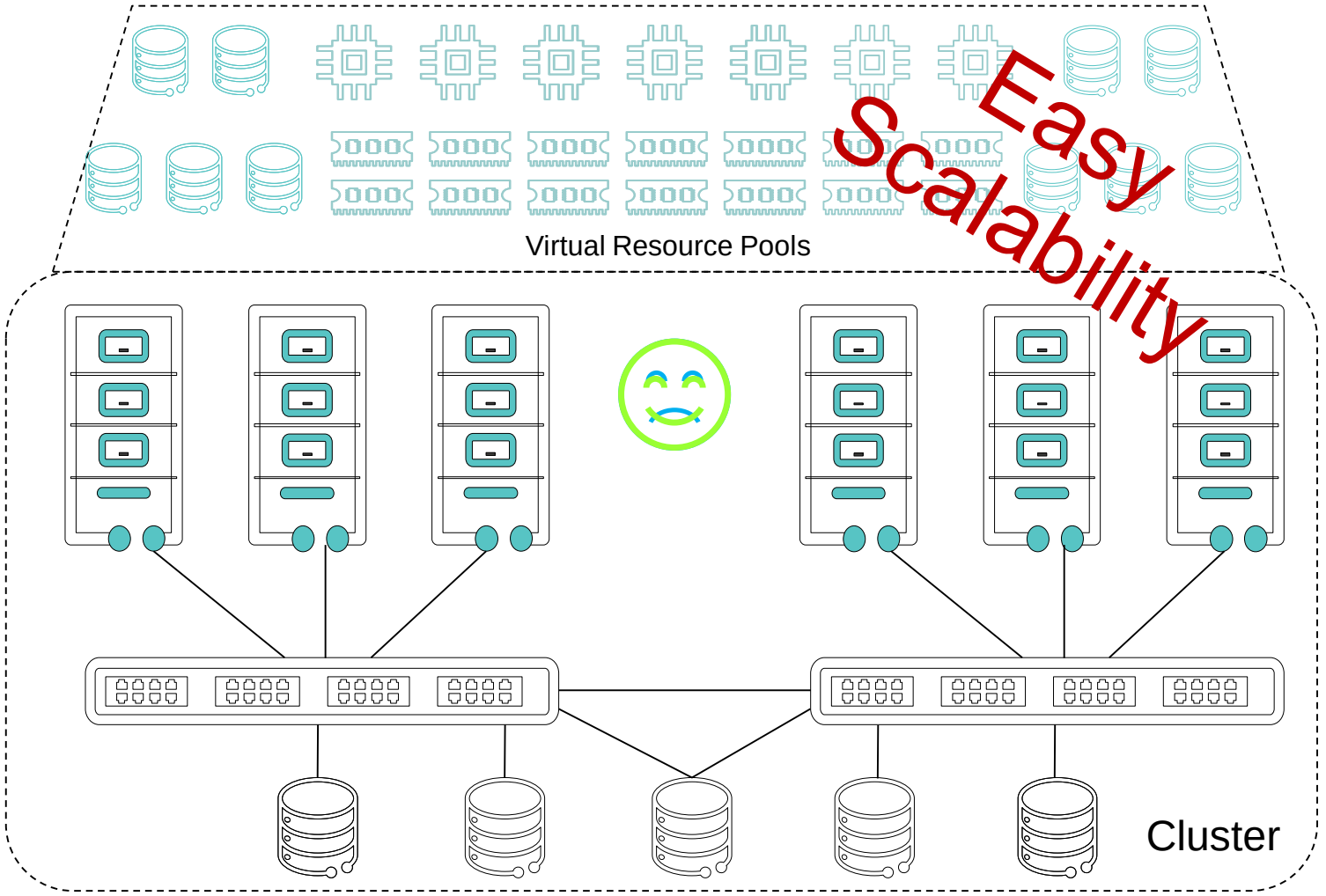
VM

VM

VM

VM

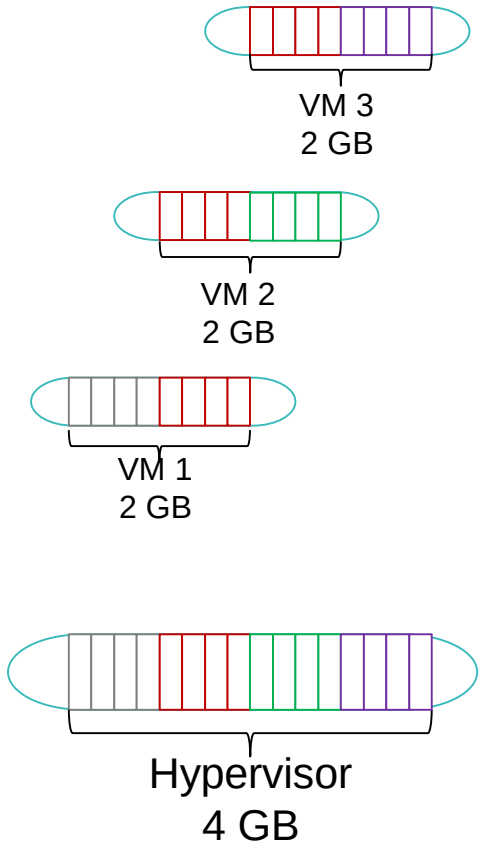
VM



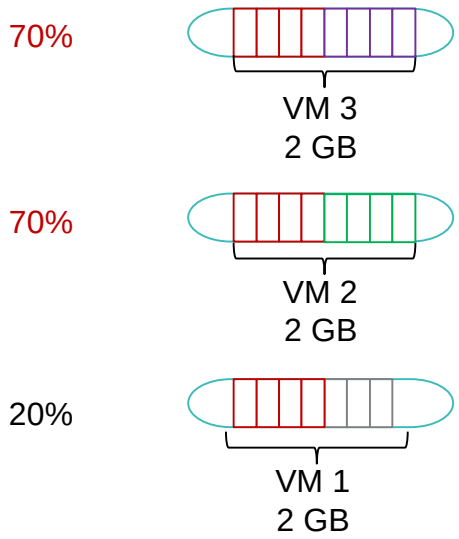
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Virtual Cluster Features (4)

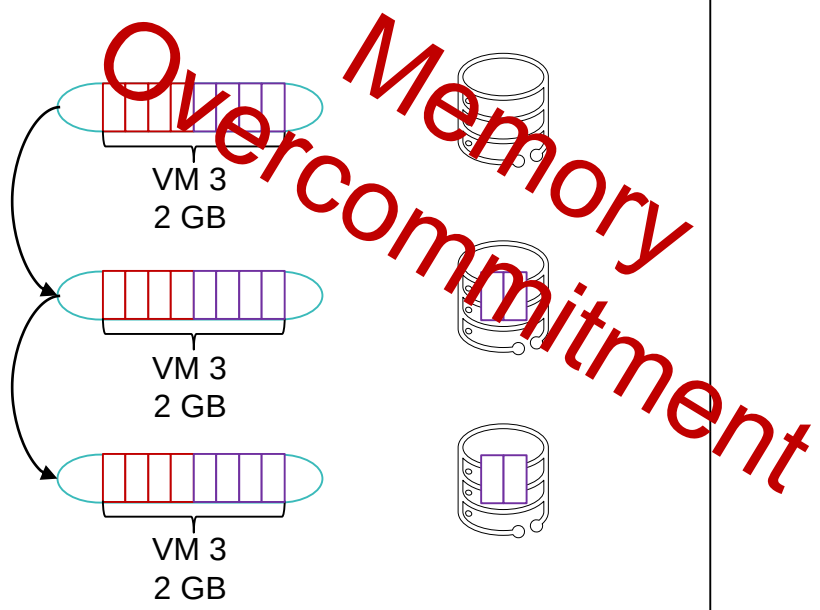
Memory Sharing



Memory Ballooning



Memory Swapping



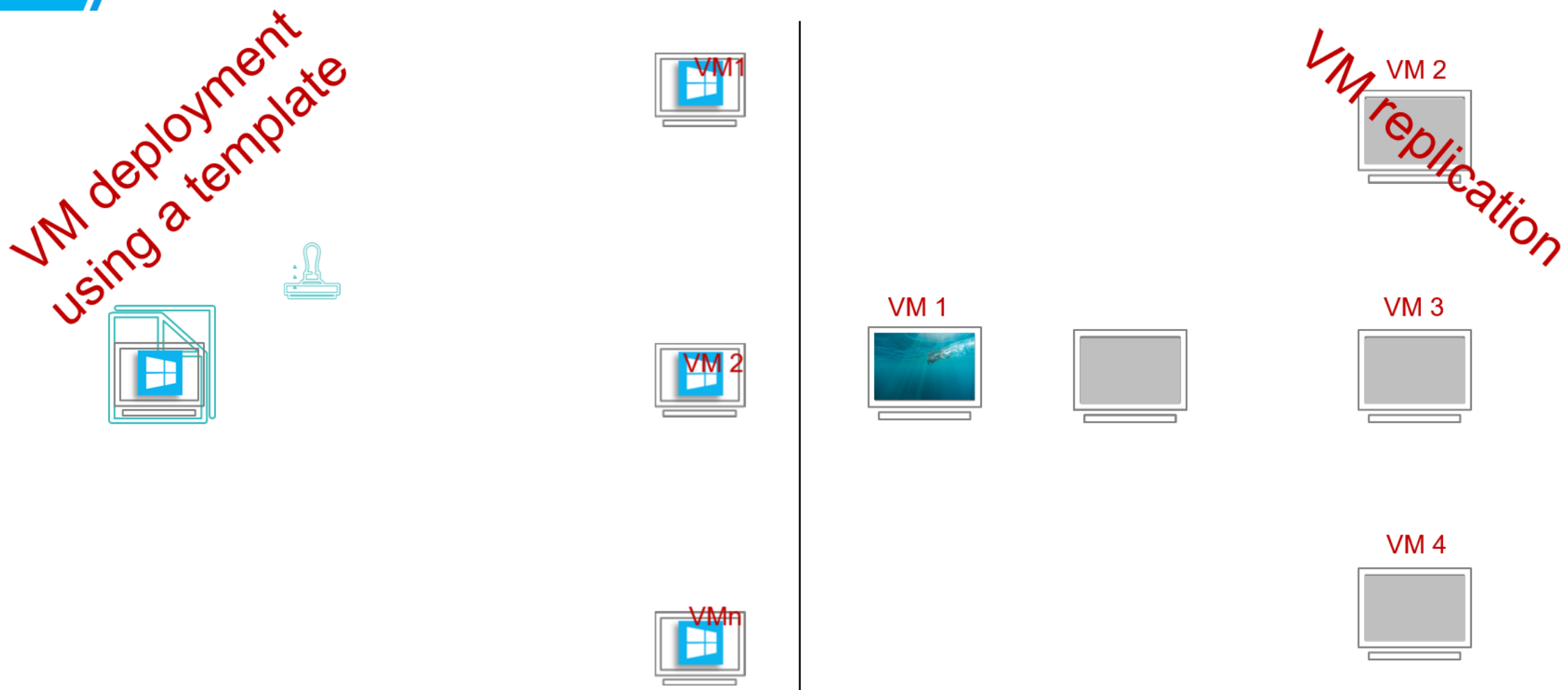
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Virtualization (VM) Features (1)

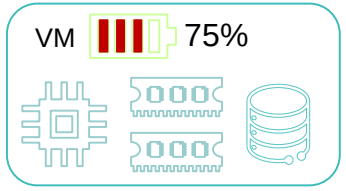


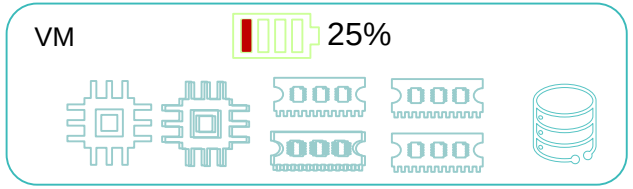
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Virtualization (VM) Features (2)

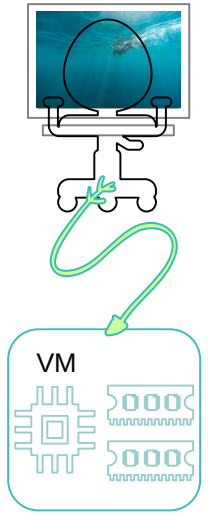
Hot-add resources





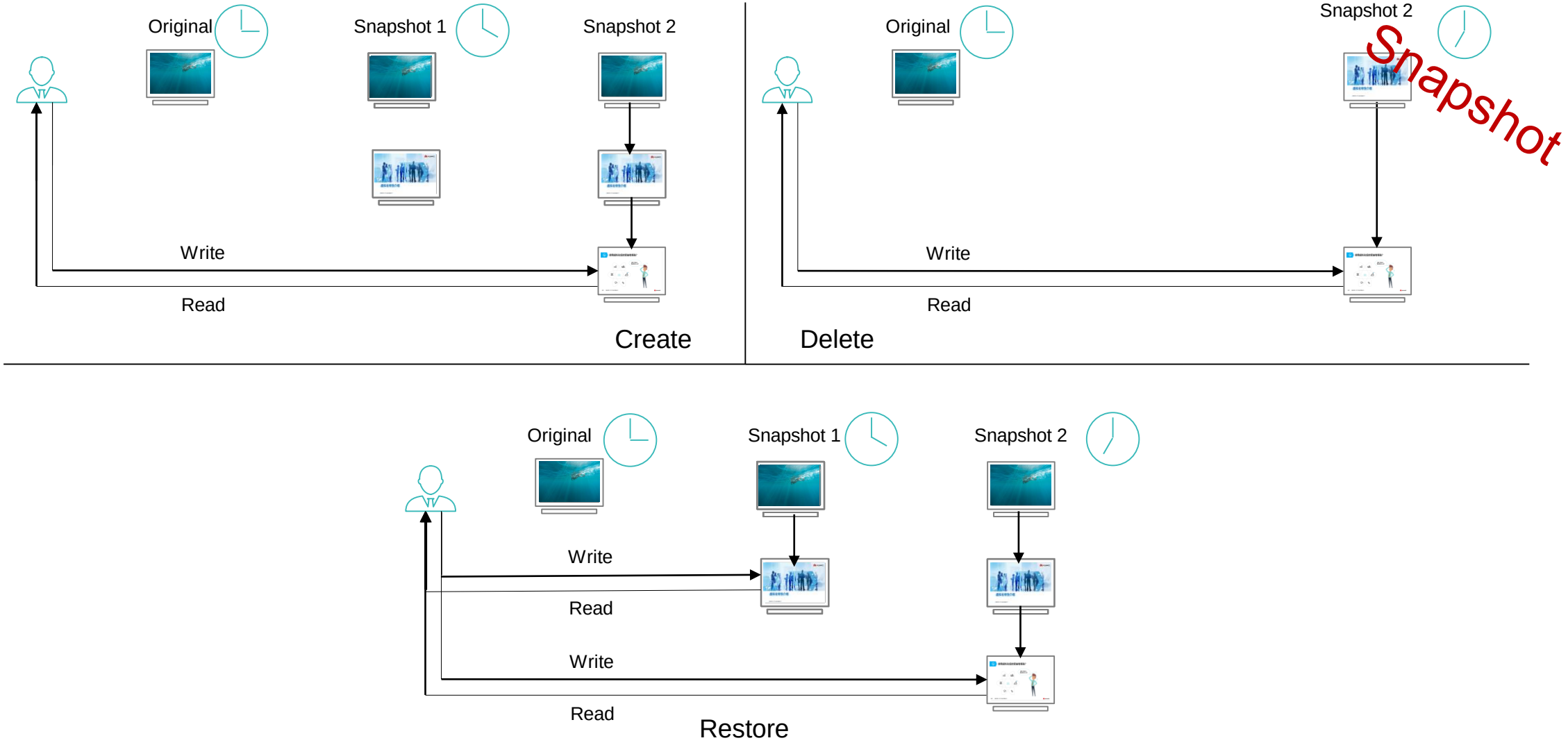


Console control



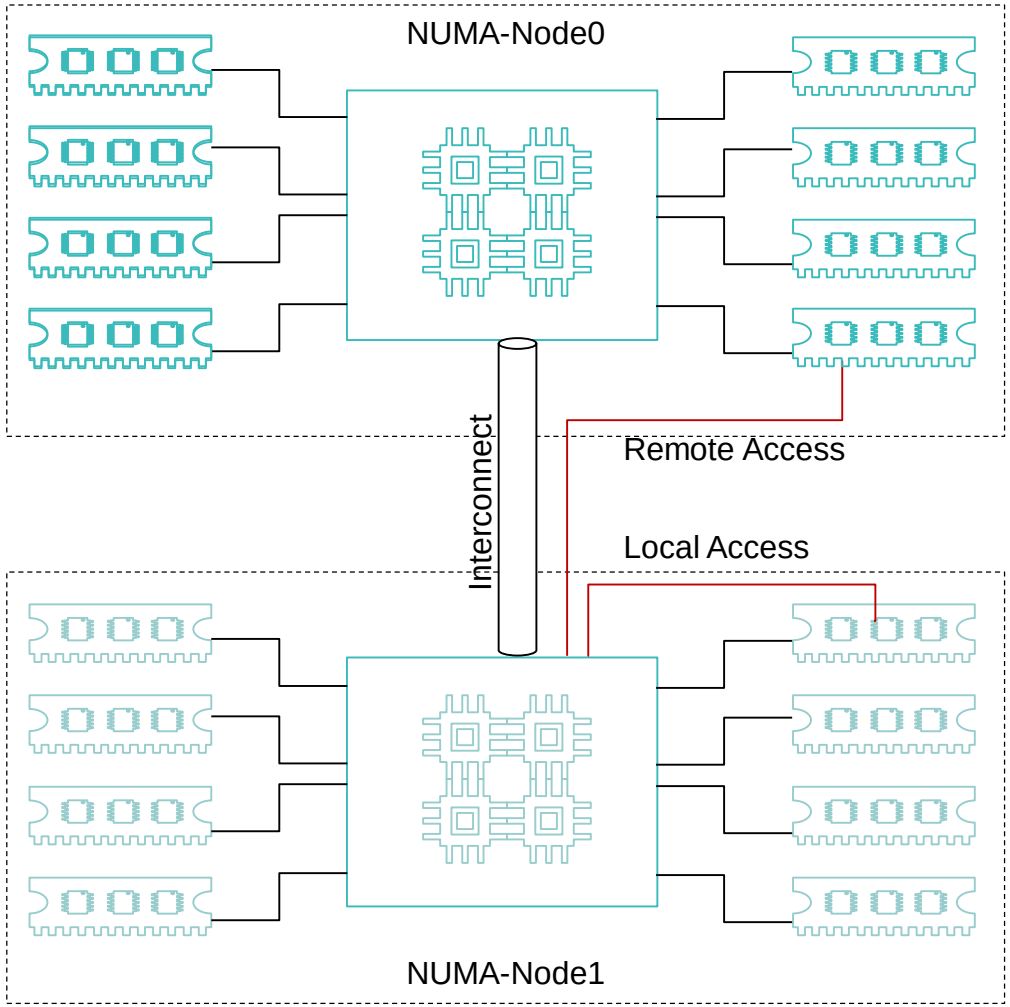
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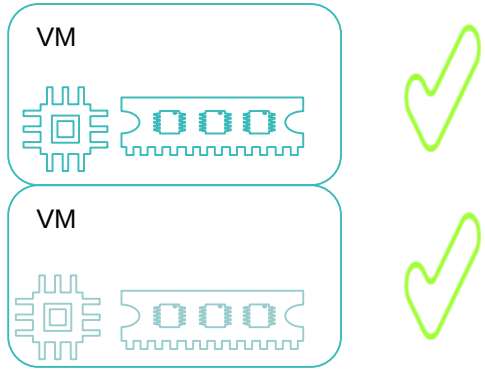
Virtualization (VM) Features (3)

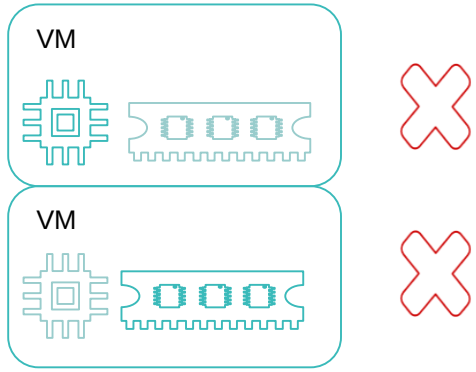


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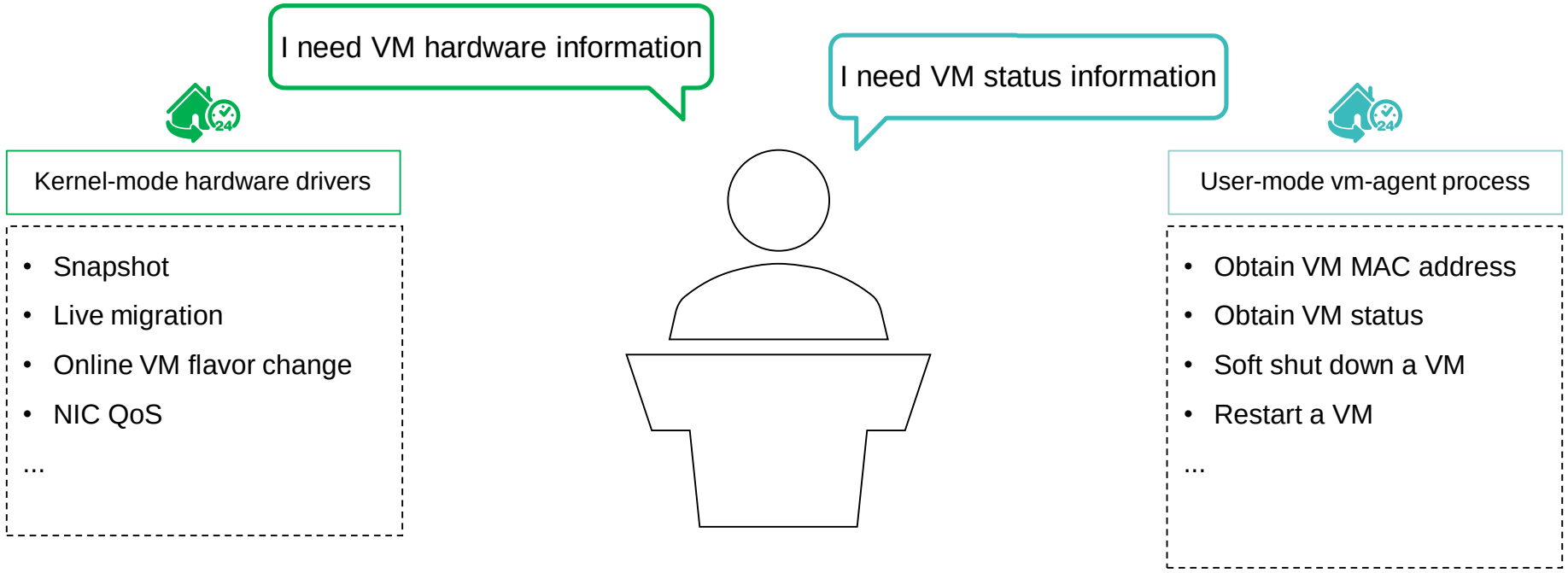
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Prerequisites for Enabling Advanced Features - Tools



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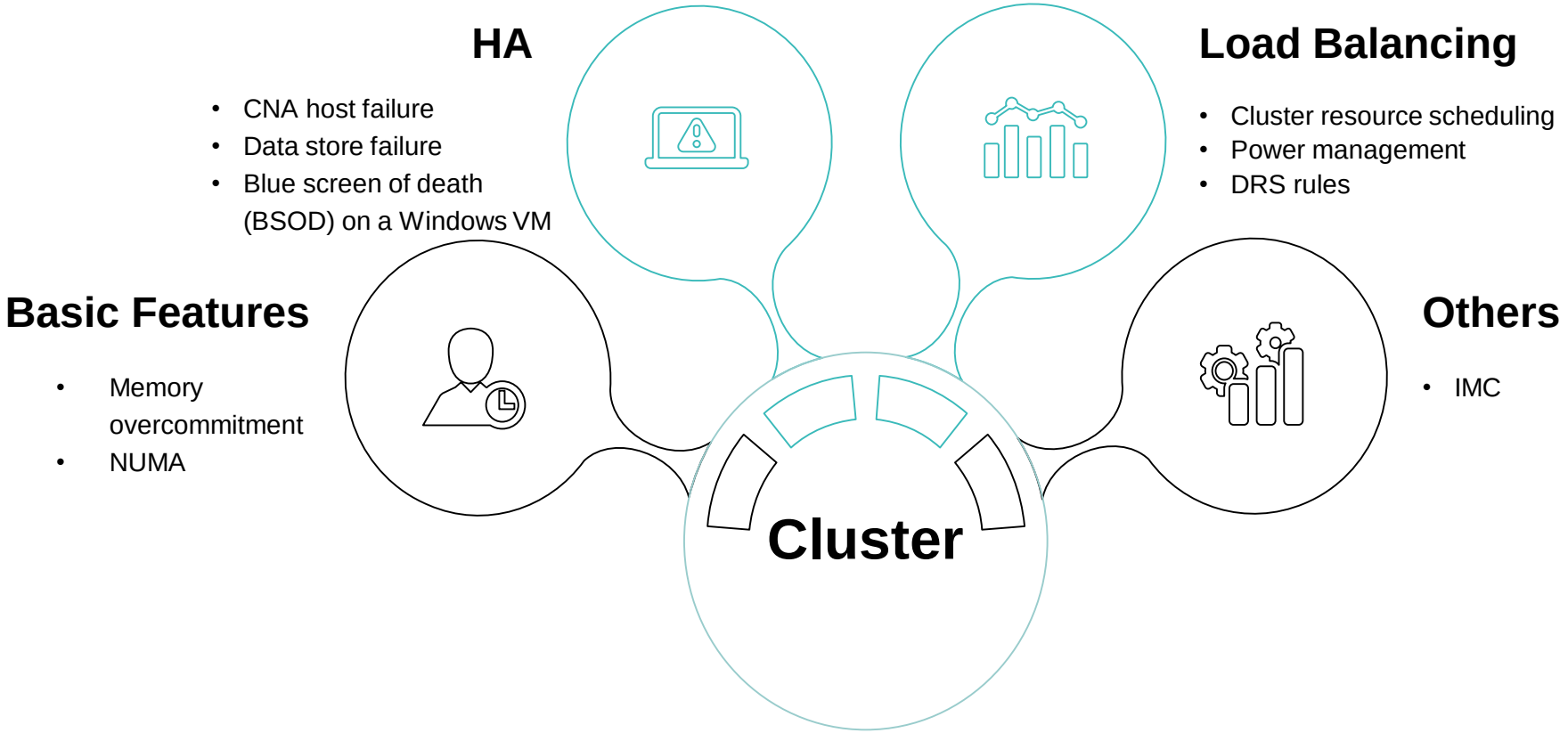
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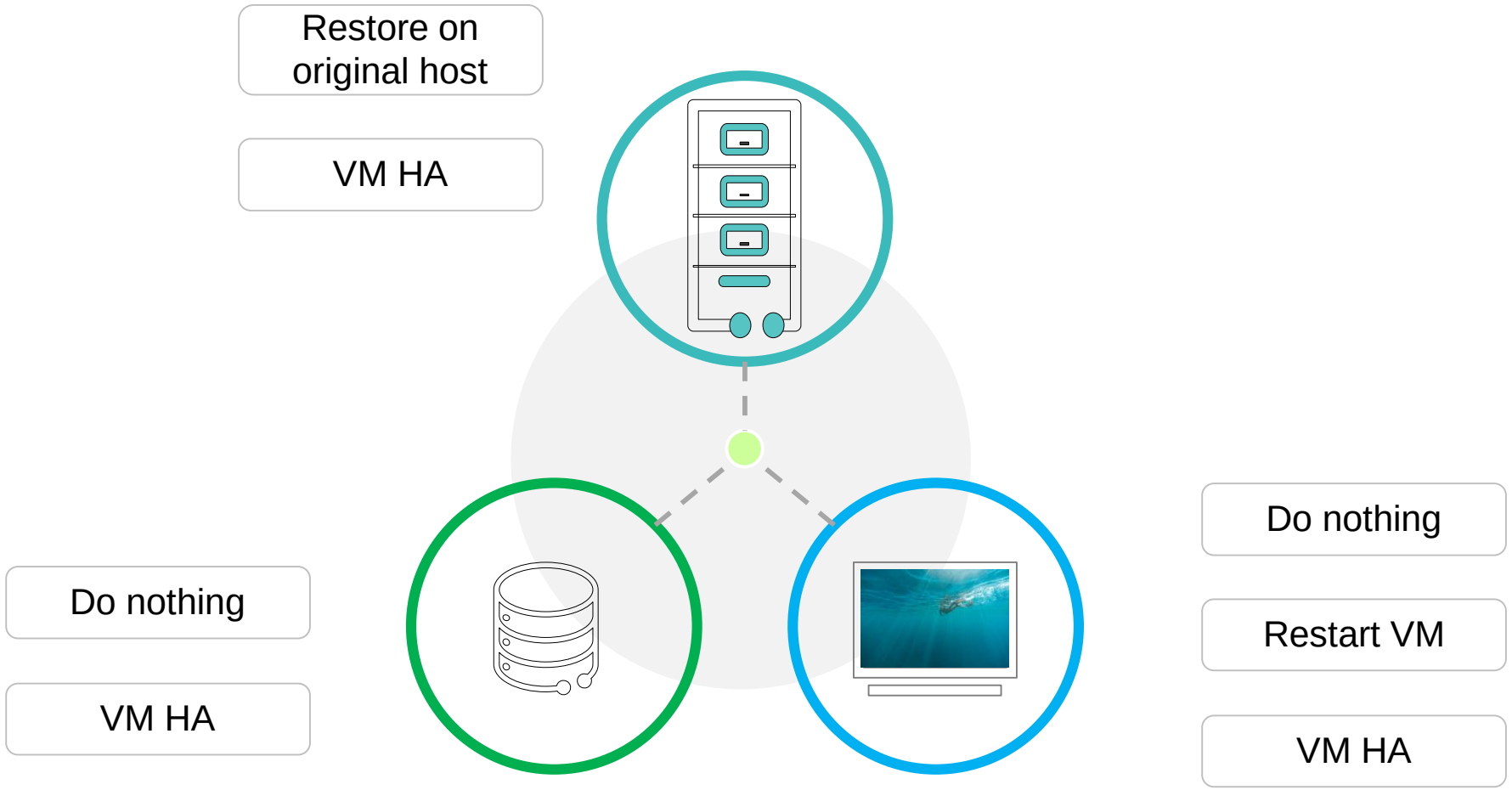
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Cluster Features



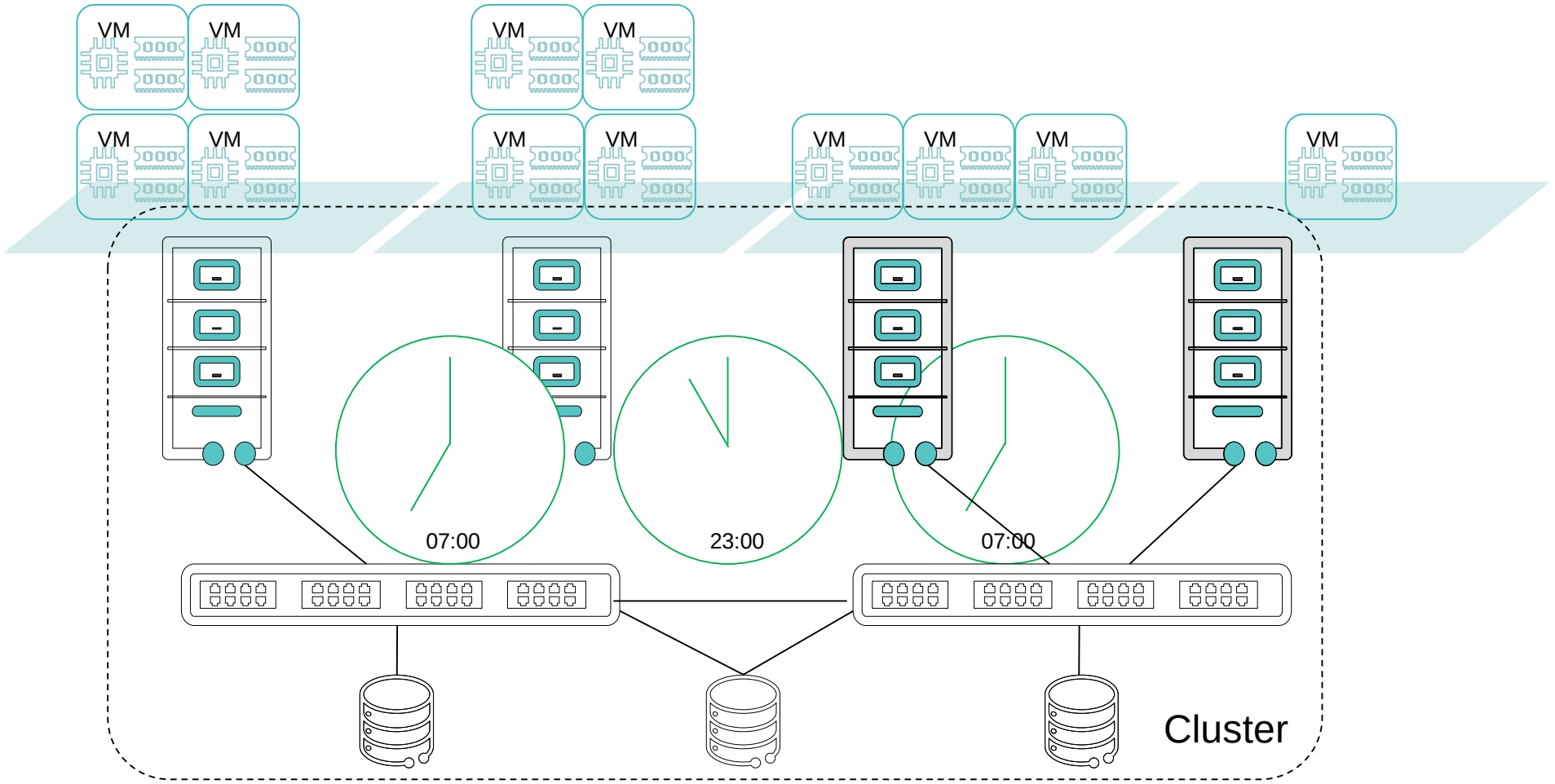
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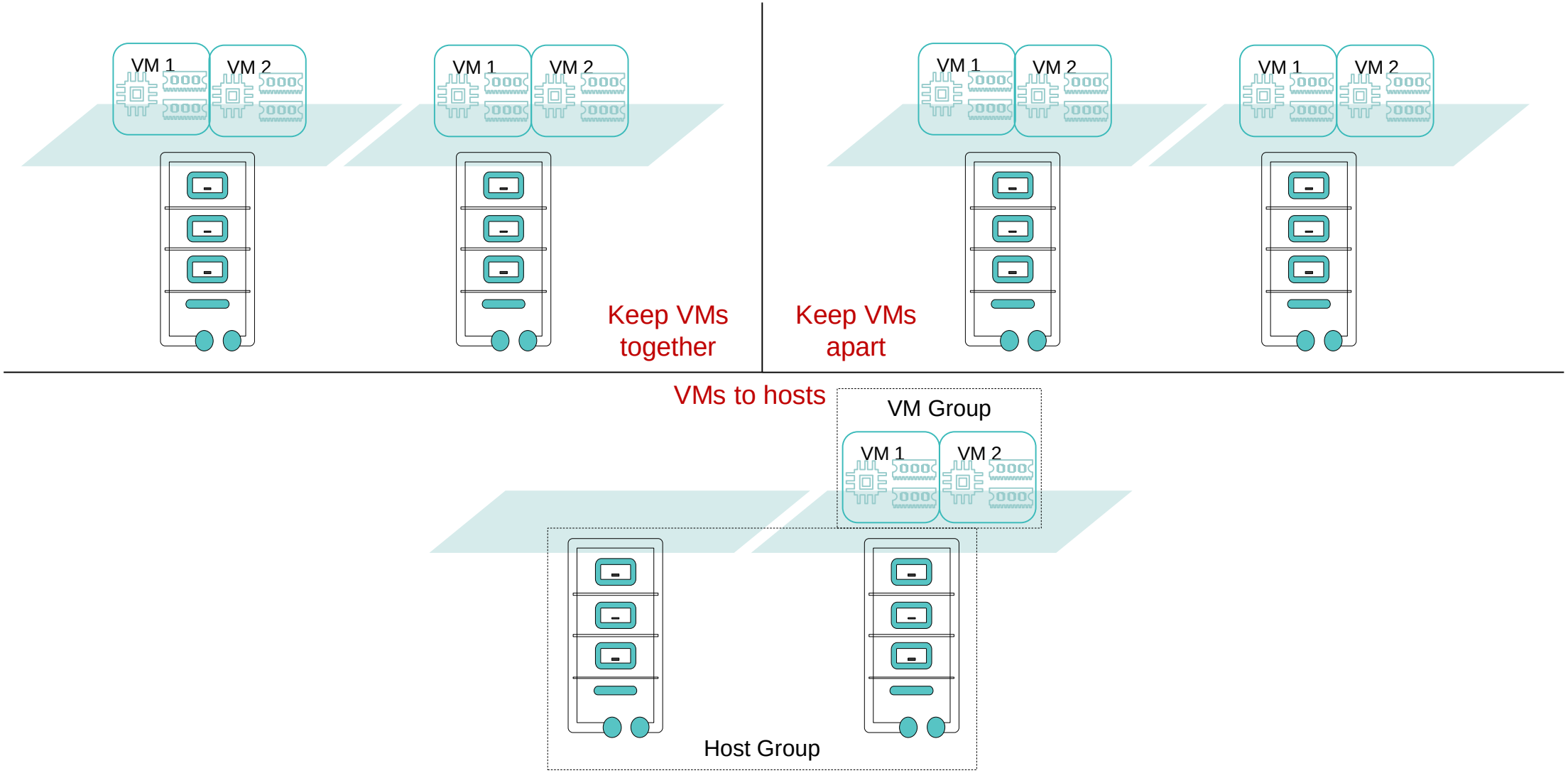
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Power Management



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DRS Rules



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IMC

- On FusionCompute, enabling incompatible migration cluster (IMC) mode for a cluster allows VMs to migrate between hosts that are using CPUs with different performance baselines within this cluster.
- Currently, IMC mode only allows VM live migration between hosts using Intel CPUs.
- IMC mode allows the hosts in a cluster to present the same CPU function set to VMs running on them, despite the fact that these hosts may be using CPUs with different performance baselines.
- For the IMC mode to work for a cluster that already contains hosts and VMs, ensure that the following conditions are met:
  - The CPU generations of the hosts in the cluster are the same as or later than the target IMC mode.
  - The CPU generations of the running or hibernating VMs in the cluster are the same as or earlier than the target IMC mode. If any VM in the cluster does not meet this requirement, stop this VM or migrate it to another cluster.

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VM Features

Resource Management

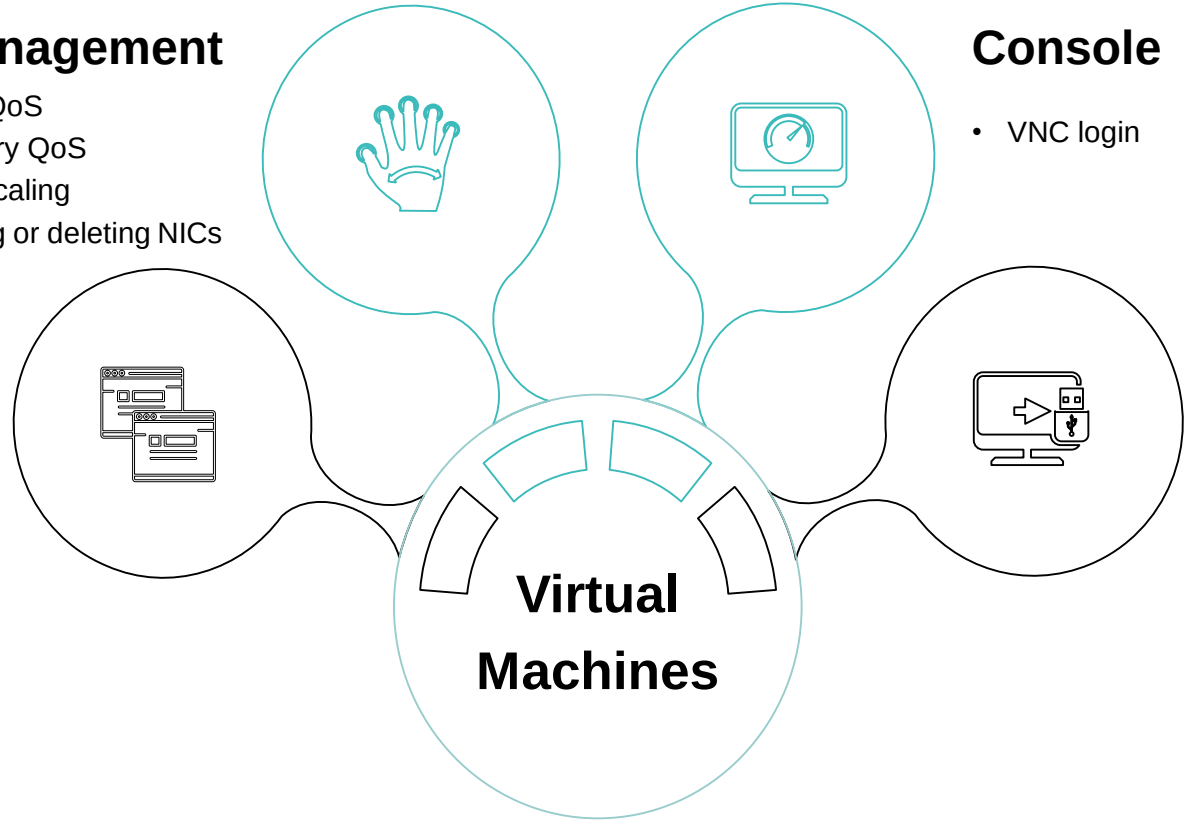
- CPU QoS
- Memory QoS
- Disk scaling
- Adding or deleting NICs

Console

- VNC login

Fast Deployment

- VM deployment using a template
- VM cloning

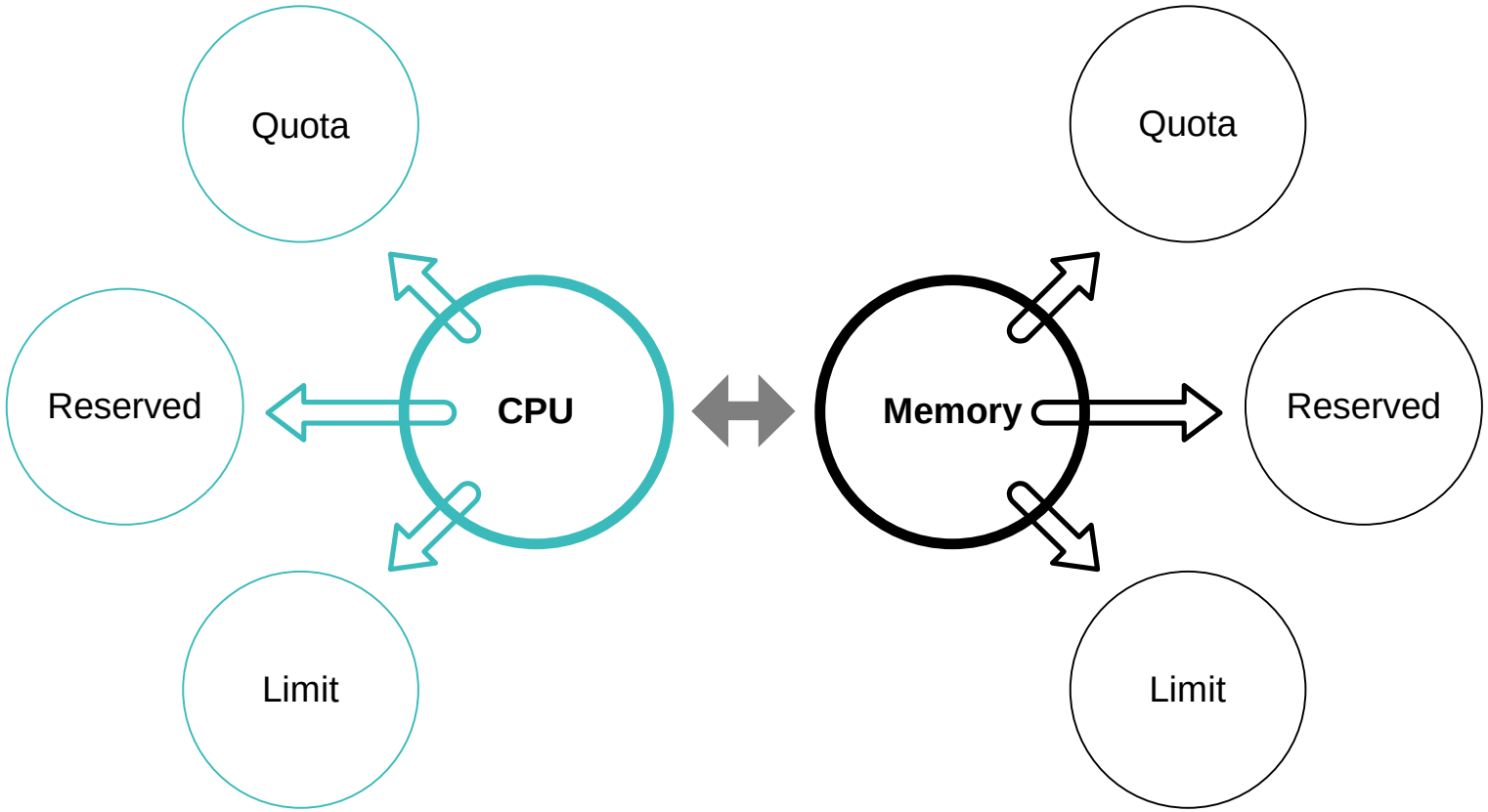


Others

- Snapshot
- Bundling or attaching USB, PCI, or GPU devices, or CD-ROM drive

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CPU and Memory QoS



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Quiz

1. Which of the following is not a snapshot management option?

   A. Create a snapshot

   B. Modify a snapshot

   C. Restore a VM using one of its snapshots

   D. Replicate a snapshot

2. Templates can be used to quickly deploy VMs. At present, templates can be created only by converting or cloning VMs.

   A. TRUE

   B. FALSE

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Summary

At this point you should:

- Understand virtualization features.
- Understand where and when virtualization features may be used.

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Recommendations

- Huawei e-Learning website
  - http://support.huawei.com/learning/Index!toTrainIndex
- Huawei Support case library
  - http://support.huawei.com/enterprise/servicecenter?lang=en
- HCIA-Cloud Computing v4.0 online forum
  - https://forum.huawei.com/enterprise/en/Huawei-Official-Communication-Channel-HCNA-Cloud-Certification-Course/thread/456287-911

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